



AgroLnk — Complete Warehouse & e-NWR Workflow Testing Manual

This manual provides an end-to-end testing blueprint for the **WDRA-Accredited e-NWR Warehousing, Multi-Chamber Cold Storage, and Commodity Collateral Financing** workflows in AgroLnk.



1. Actors & Roles in the Warehouse Ecosystem

Role	Test User Email	Password	Primary Workflow Actions		:---	:---	:---	:---
Warehouse Operator	`warehouse@agrolnk.com`	`pass123` (or any)	View multi-chamber occupancy, audit storage receipts, inspect temperature zones, manage inward/outward gatepasses.					
Verified Farmer	`farmer@agrolnk.com`	`farmer123`	Deposit harvested produce, receive digital **e-NWR titles**, apply for 75% LTV commodity loans, list for sale directly from storage.					
Financier / NBFC	`financier@agrolnk.com`	`pass123`	Underwrite working capital against e-NWR receipts, place collateral liens in the **Collateral Vault**, disburse credit.					
Wholesale Buyer	`buyer@agrolnk.com`	`buyer123`	Procure warehouse-stored produce with verified assay certificates and immediate delivery/title transfer.					



2. Warehouse Life-Cycle Architecture

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` [ 1. Inward Physical Deposit ] [ 2. e-NWR Title Issuance ] (Assay, Moisture, Chamber) / \ / \ [ 3. Collateral Pledge / Loan ] [ 4. Sell from Storage ] (75% LTV Working Capital) (Spot Market / Auction) \ / \ / [ 5. Outward Delivery / Release ]`
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3. Master Step-by-Step Test Scenarios

◆ SCENARIO 1: Farmer Inward Deposit & e-NWR Receipt Creation

Objective: Validate that a farmer can choose an accredited warehouse facility, specify produce assay parameters, calculate monthly storage costs, and generate a WDRA electronic Negotiable Warehouse Receipt (e-NWR).

Steps for Tester:

1. Login as Farmer (farmer@agroInk.com). 2. Navigate to Warehouse & e-NWR Vault (#/farmer-inventory or click Inventory in the sidebar). 3. Click the + Deposit Produce button in the header banner. 4. In the Deposit to Accredited Warehouse Modal: * Select Warehouse: Salem Agri Cold Storage Hub (WDRA/2024/TN/0892) * Commodity: Tomato * Variety: Hybrid Shivam * Grade: Grade A Premium * Quantity: 1000 kg (1.0 MT) * Estimated Value: ₹42 / kg (₹42,000 Total Valuation) * Assigned Chamber: Chamber B4 (Ultra Cold: 2°C - 4°C) * Storage Duration: 60 Days 5. Verify Cost Estimation Display: * Observe the automated monthly storage fee calculation: 1.0 MT x ₹350/MT = ₹350 / month . 6. Click Confirm & Generate e-NWR Receipt.

• Expected Outcome:

* A new electronic receipt (e.g. eNWR-2024-8841) is generated instantly. * The receipt is added to the farmer's active inventory with status Stored . * Total stored metric and valuation counters update automatically.

◆ SCENARIO 2: Inspecting e-NWR Title Certificate & Assay Parameters

Objective: Confirm that the generated e-NWR contains full legal title ownership, QR/WDRA accreditation, and quality assay data.

Steps for Tester:

1. On the Farmer Inventory page (#/farmer-inventory), locate the newly created receipt card. 2. Click View e-NWR Certificate. 3. Verify the Electronic Title Modal: * Title Header: Electronic Negotiable Warehouse Receipt (WDRA Accredited Title) * Receipt ID: Formatted as eNWR-2024-XXXX with an emerald badge. * Chamber & Temperature Specs: Displays exact room temp (e.g., Chamber B4 (2°C - 4°C)). * Assay Parameters: Moisture %, foreign matter %, and grade classification. * Action CTAs Available: * [🏪 Sell from Storage] * [🏠]

Apply for Working Capital]

- **Expected Outcome:** High-fidelity, bankable electronic title document viewable on any device.

◆ SCENARIO 3: Warehouse Operator Dashboard & Multi-Chamber Auditing

Objective: Verify that the warehouse operator has complete oversight of capacity, chamber temperatures, and stored receipts.

Steps for Tester:

1. Login as Warehouse Operator (warehouse@agroInk.com). 2. Land on the Warehouse Management & e-NWR Terminal (#/warehouse-dashboard). 3. **Verify Metric Cards:** * **Total Capacity:** 5,000 Tonnes * **Occupancy %:** Dynamically calculated based on active stored tonnes. * **Active e-NWRs:** Count of valid receipts in custody. * **Total Custody Valuation:** Aggregated rupee valuation of all stored commodities. 4. **Switch to Chamber Overview Tab:** * Verify all 4 temperature zones: * Chamber A1 (Dry Storage) – Ambient (24°C) (Turmeric, Grains) * Chamber B2 (Cold Cell) – 4°C – 8°C (Potatoes, Carrots) * Chamber B4 (Ultra Cold) – 2°C – 4°C (Hybrid Tomatoes, Fruits) * Chamber C1 (CA Controlled) – 0°C – 2°C (Export Apples, Grapes) 5. **Switch to Inventory & Receipts Tab:** * Search for the farmer's receipt eNWR-2024-XXXX . * Click **Audit e-NWR** to review physical custody specs.

- **Expected Outcome:** Real-time visibility into multi-chamber utilization and commodity custody.

◆ SCENARIO 4: 75% LTV Working Capital Loan against e-NWR

Objective: Validate that farmers can pledge stored commodities to unlock instant working capital without selling early.

Steps for Tester:

1. Login as Farmer (farmer@agroInk.com) and go to e-NWR Vault (#/farmer-inventory). 2. On any stored receipt card, click **Apply for Working Capital** (or open the receipt detail modal and click **Apply for Working Capital**). 3. In the **e-NWR Collateral Financing Modal:** * **Selected Commodity & Valuation:** ₹42,000 * **Maximum Loan Facility (75% LTV):** ₹31,500 * **Enter Requested Amount:** ₹25,000 * **Select Tenor:** 30 Days (1.2% monthly interest) * **Note:** Pre-harvest liquidity

for input fertilizers. 4. Click **Submit Working Capital Request**.

- **Expected Outcome:** Financing request is routed directly to the Financier / NBFC underwriting queue.
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◆ SCENARIO 5: NBFC Credit Approval & Collateral Vault Lien

Objective: Confirm that the financier reviews the warehouse receipt, places a legal lien on the commodity, and disburses credit.

Steps for Tester:

1. Login as Financier / NBFC (`financier@agroInk.com`). 2. Go to **Financier Portal** → **Underwriting Desk** (`#/financier-underwriting`). 3. Locate the submitted loan application for the farmer (₹25,000 backed by eNWR-2024-XXXX). 4. Click **Review & Underwrite**: * Inspect WDRA warehouse accreditation code. * Inspect commodity valuation buffer. 5. Click **Approve & Disburse Funds**. 6. **Navigate to Collateral Vault** (`#/financier-collateral-vault`): * Verify that the receipt is now listed in the **Collateral Vault** with an active **Lien / Pledge Badge**. * Commodity quantity is locked from unauthorized farmer withdrawal.

- **Expected Outcome:** Seamless institutional credit underwriting backed by digital agri-collateral.
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◆ SCENARIO 6: Listing for Sale Directly from Warehouse (Zero Transit Latency)

Objective: Verify that farmers can list all or part of their stored commodity on the Spot Marketplace or Live Auction directly from the warehouse.

Steps for Tester:

1. Login as Farmer (`farmer@agroInk.com`) and go to **e-NWR Vault** (`#/farmer-inventory`). 2. On the receipt card, click **Sell from Storage**. 3. In the **List from Warehouse Modal**: * Choose Sale Type: **Direct Spot Listing (or Live Auction)** * Available Stored: 1000 kg * Quantity to List: 500 kg (Partial listing test) * Selling Price: ₹45/kg 4. Click **Publish to Marketplace**. 5. **Verify System Updates**: * The receipt's available quantity decreases to 500 kg, and status reflects `partially_listed`. * Login as Buyer (`buyer@agroInk.com`) and open **Marketplace** (`#/buyer-marketplace`). * The newly published lot appears on the marketplace marked with a  **Warehouse-Stored (WDRA Certified)** badge, enabling

buyers to purchase pre-stored, pre-assayed goods with zero delay.

- **Expected Outcome:** Frictionless transition from storage into digital commerce without double handling or freight wastage.

4. Warehouse & e-NWR QA Verification Checklist

Test Item	Verification Criteria	Status		---	---	---																						
Accredited Facility Selection	Only WDRA verified hubs appear in the deposit selector.		Pass	Storage Fee Formula	(Weight in Tonnes) × (Rate/Tonne/Month) × (Months)	calculated accurately.		Pass	e-NWR Receipt Numbering	Clean sequential/timestamped format (eNWR-YYYY-XXXX`).		Pass	Temperature Zone Mapping	Produce types correctly mapped to designated cold chain chambers.		Pass	Partial Quantity Listing	Stored inventory decrements correctly when a portion is listed on the marketplace.		Pass	Collateral Lien Protection	Financier pledge locks receipt from being listed or withdrawn until loan settlement.		Pass	Multi-Role Synchronization	Updates made by Farmer/Operator/Financier reflect in real time.		Pass